

## Environment Setup

```
Command Prompt - node ser X Command Prompt - node ser X + v
C:\Users\Vida Reddy\ecommerce-microservices>python --version
Python 3.14.4

C:\Users\Vida Reddy\ecommerce-microservices>pip --version
pip 26.1.2 from C:\Users\Vida Reddy\AppData\Roaming\Python\Python314\site-packages\pip (python 3.14)

C:\Users\Vida Reddy\ecommerce-microservices>node --version
v20.20.2

C:\Users\Vida Reddy\ecommerce-microservices>npm --version
10.8.2

C:\Users\Vida Reddy\ecommerce-microservices>mongosh --version
2.12.0

C:\Users\Vida Reddy\ecommerce-microservices>mkdir user-service

C:\Users\Vida Reddy\ecommerce-microservices>cd user-service

C:\Users\Vida Reddy\ecommerce-microservices\user-service>npm init -y
Wrote to C:\Users\Vida Reddy\ecommerce-microservices\user-service\package.json:

{
  "name": "user-service",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
```

## User Service Setup

```
Command Prompt - node ser X Command Prompt - node ser X + v
C:\Users\Vida Reddy\ecommerce-microservices\user-service>npm install express mongoose dotenv bcryptjs jsonwebtoken cors
added 104 packages, and audited 105 packages in 14s

32 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\Vida Reddy\ecommerce-microservices\user-service>notepad .env

C:\Users\Vida Reddy\ecommerce-microservices\user-service>mkdir models

C:\Users\Vida Reddy\ecommerce-microservices\user-service>notepad models\User.js

C:\Users\Vida Reddy\ecommerce-microservices\user-service>dir models
Volume in drive C is Windows-SSD
Volume Serial Number is F2F9-98BB

Directory of C:\Users\Vida Reddy\ecommerce-microservices\user-service\models

29-09-2026  09:31    <DIR>          .
29-09-2026  09:31    <DIR>          ..
29-09-2026  09:31                371 User.js
               1 File(s)                371 bytes
               2 Dir(s)  132,570,509,312 bytes free

C:\Users\Vida Reddy\ecommerce-microservices\user-service>mkdir controllers

C:\Users\Vida Reddy\ecommerce-microservices\user-service>notepad controllers\userController.js

C:\Users\Vida Reddy\ecommerce-microservices\user-service>dir controllers
Volume in drive C is Windows-SSD
Volume Serial Number is F2F9-98BB

Directory of C:\Users\Vida Reddy\ecommerce-microservices\user-service\controllers
```

## User Service Running

```
Command Prompt - node ser X Command Prompt - node ser X + v
Volume in drive C is Windows-SSD
Volume Serial Number is F2F9-98BB

Directory of C:\Users\Vida Reddy\ecommerce-microservices\user-service\controllers

29-09-2026 09:32 <DIR> .
29-09-2026 09:32 <DIR> ..
29-09-2026 09:36      1,793 userController.js
               1 File(s)      1,793 bytes
               2 Dir(s)  132,595,773,440 bytes free

C:\Users\Vida Reddy\ecommerce-microservices\user-service>mkdir routes

C:\Users\Vida Reddy\ecommerce-microservices\user-service>notepad routes\userRoutes.js

C:\Users\Vida Reddy\ecommerce-microservices\user-service>dir routes
Volume in drive C is Windows-SSD
Volume Serial Number is F2F9-98BB

Directory of C:\Users\Vida Reddy\ecommerce-microservices\user-service\routes

29-09-2026 09:37 <DIR> .
29-09-2026 09:37 <DIR> ..
29-09-2026 09:38      298 userRoutes.js
               1 File(s)      298 bytes
               2 Dir(s)  132,563,320,832 bytes free

C:\Users\Vida Reddy\ecommerce-microservices\user-service>notepad server.js

C:\Users\Vida Reddy\ecommerce-microservices\user-service>node server.js
◇ injected env (3) from .env
MongoDB connected
User Service running on port 4001
|
```

## Activity Service Setup

```
Command Prompt - node ser X Command Prompt - node ser X + v
Microsoft Windows [Version 10.0.26200.9550]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Vida Reddy>

C:\Users\Vida Reddy>cd "C:\Users\Vida Reddy\ecommerce-microservices"

C:\Users\Vida Reddy\ecommerce-microservices>mkdir activity-service

C:\Users\Vida Reddy\ecommerce-microservices>cd activity-services
The system cannot find the path specified.

C:\Users\Vida Reddy\ecommerce-microservices>cd activity-service

C:\Users\Vida Reddy\ecommerce-microservices\activity-service>npm init -y
Wrote to C:\Users\Vida Reddy\ecommerce-microservices\activity-service\package.json:

{
  "name": "activity-service",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}

C:\Users\Vida Reddy\ecommerce-microservices\activity-service>npm install express mongoose dotenv cors
added 89 packages, and audited 90 packages in 10s
```

## Activity Service Running

```
Command Prompt - node ser x Command Prompt - node ser x + v

C:\Users\Vida Reddy\ecommerce-microservices\activity-service>npm install express mongoose doter
added 89 packages, and audited 90 packages in 10s

31 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\Vida Reddy\ecommerce-microservices\activity-service>notepad .env

C:\Users\Vida Reddy\ecommerce-microservices\activity-service>mkdir models

C:\Users\Vida Reddy\ecommerce-microservices\activity-service>notepad models\Activity.js

C:\Users\Vida Reddy\ecommerce-microservices\activity-service>notepad server.js

C:\Users\Vida Reddy\ecommerce-microservices\activity-service>node server.js
◇ injected env (2) from .env
MongoDB connected
Activity Service running on port 4002
```

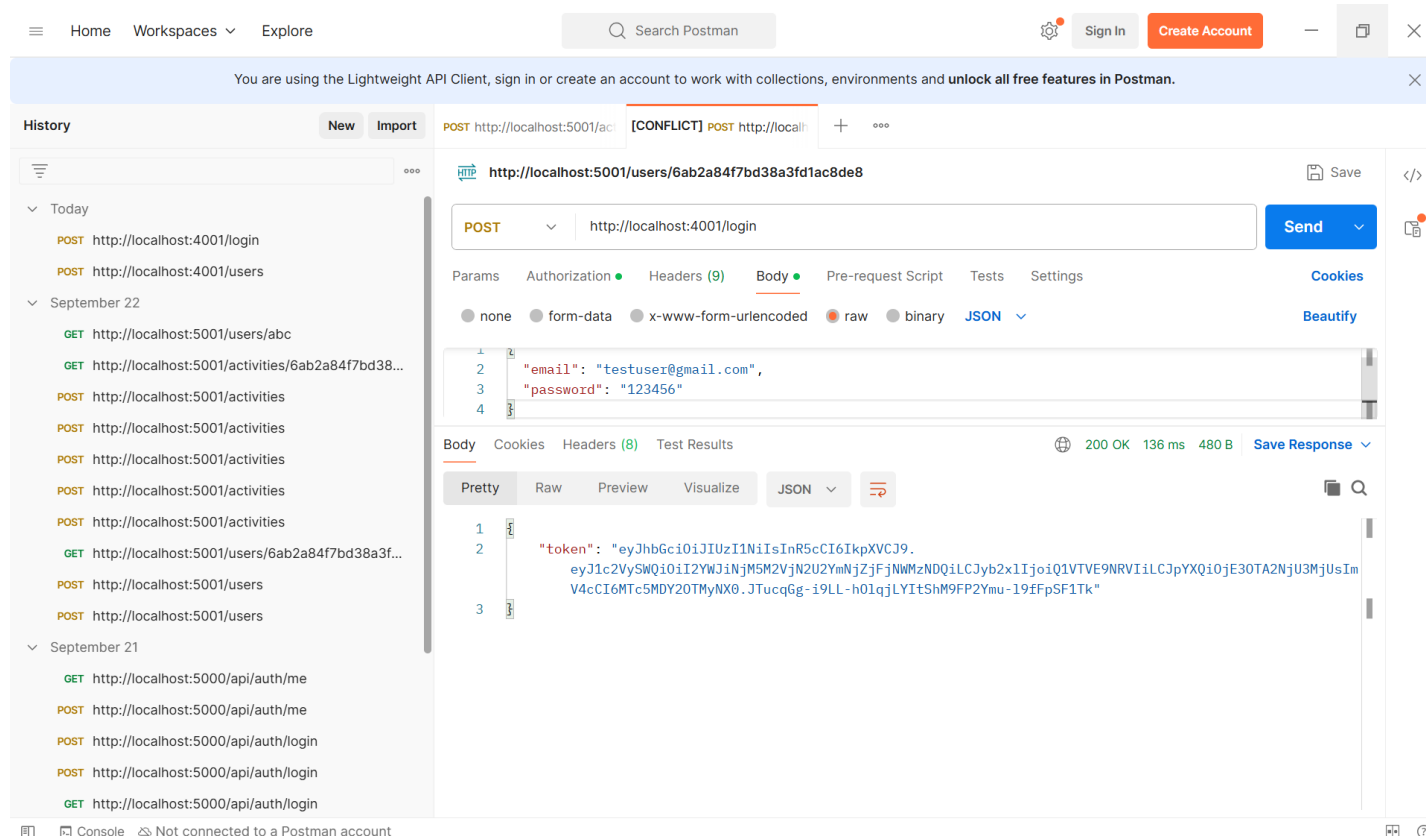
## User Registration API

The screenshot displays the Postman interface with a successful POST request to the User Registration API. The URL is `http://localhost:5001/users/6ab2a84f7bd38a3fd1ac8de8`. The request body is a JSON object containing user details: `{ "name": "Test User", "email": "testuser@gmail.com", "password": "123456" }`. The response is a JSON object with the following fields: `{ "id": "6abb6393ec7e6bccf1c5c344", "name": "Test User", "email": "testuser@gmail.com", "role": "CUSTOMER" }`. The status code is 201 Created, and the response time is 663 ms.

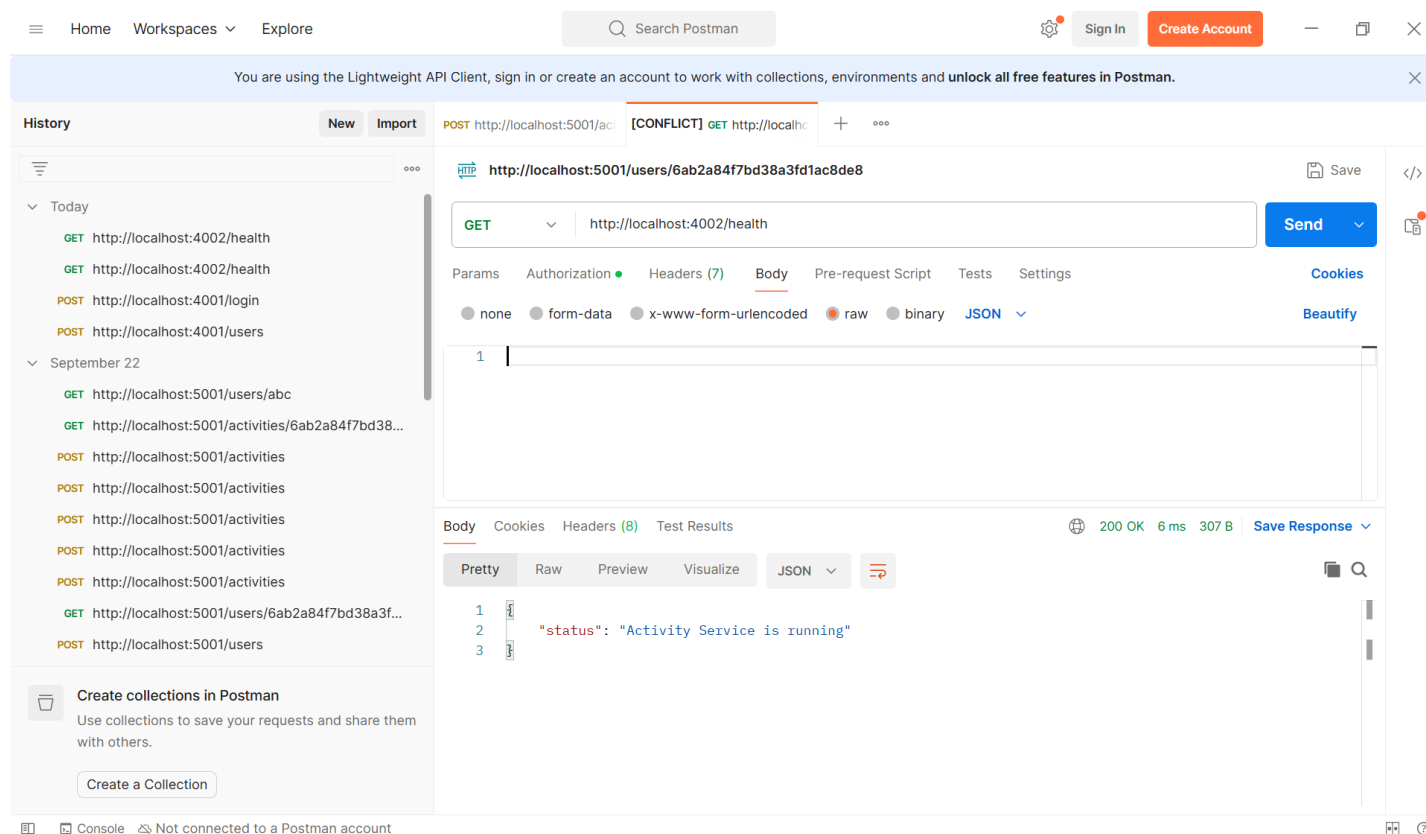
History: Today  
POST http://localhost:4001/users  
September 22  
GET http://localhost:5001/users/abc  
GET http://localhost:5001/activities/6ab2a84f7bd38a3fd1ac8de8  
POST http://localhost:5001/activities  
POST http://localhost:5001/activities  
POST http://localhost:5001/activities  
POST http://localhost:5001/activities  
POST http://localhost:5001/activities  
GET http://localhost:5001/users/6ab2a84f7bd38a3fd1ac8de8  
POST http://localhost:5001/users  
POST http://localhost:5001/users  
September 21  
GET http://localhost:5000/api/auth/me  
POST http://localhost:5000/api/auth/me  
POST http://localhost:5000/api/auth/login  
POST http://localhost:5000/api/auth/login  
GET http://localhost:5000/api/auth/login  
September 18

Console: Not connected to a Postman account

## User Login API



## Activity Service Health Check



## Activity Creation API

HomeWorkspacesExplore

Search Postman

Sign InCreate Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History

NewImport

POST http://localhost:5001/ac[CONFLICT] GET http://localho

Today

GET http://localhost:4002/healthGET http://localhost:4002/healthPOST http://localhost:4001/loginPOST http://localhost:4001/users

September 22

GET http://localhost:5001/users/abcGET http://localhost:5001/activities/6ab2a84f7bd38a3f...POST http://localhost:5001/activitiesPOST http://localhost:5001/activitiesPOST http://localhost:5001/activitiesPOST http://localhost:5001/activitiesPOST http://localhost:5001/activitiesGET http://localhost:5001/users/6ab2a84f7bd38a3f...POST http://localhost:5001/users

Create collections in Postman

Use collections to save your requests and share them with others.

Create a Collection

HTTP http://localhost:5001/users/6ab2a84f7bd38a3fd1ac8de8

GET http://localhost:4002/activities

ParamsAuthorizationHeaders (9)BodyPre-request ScriptTestsSettings

noneform-datax-www-form-urlencoderauwbinaryJSON

Save Beautify

12345

"userId": "6abb6393ec7e6bccf1c5c344",  
"action": "LOGIN",  
"description": "User logged in"

BodyCookiesHeaders (8)Test Results

200 OK6 ms307 B

Save Response

PrettyRawPreviewVisualizeJSON

"status": "Activity Service is running"

ConsoleNot connected to a Postman account

## Activity Retrieval API

HomeWorkspacesExplore

Search Postman

Sign InCreate Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History

NewImport

POST http://localhost:5001/ac[CONFLICT] GET http://localho

Today

GET http://localhost:4002/activitiesGET http://localhost:4002/activitiesPOST http://localhost:4002/activitiesGET http://localhost:4002/activitiesGET http://localhost:4002/healthPOST http://localhost:4001/loginPOST http://localhost:4001/users

September 22

GET http://localhost:5001/users/abcGET http://localhost:5001/activities/6ab2a84f7bd38a3f...POST http://localhost:5001/activitiesPOST http://localhost:5001/activitiesPOST http://localhost:5001/activities

Create collections in Postman

Use collections to save your requests and share them with others.

Create a Collection

HTTP http://localhost:5001/users/6ab2a84f7bd38a3fd1ac8de8

GET http://localhost:4002/activities

ParamsAuthorizationHeaders (7)BodyPre-request ScriptTestsSettings

noneform-datax-www-form-urlencoderauwbinaryJSON

Save Beautify

1

BodyCookiesHeaders (8)Test Results

200 OK7 ms487 B

Save Response

PrettyRawPreviewVisualizeJSON

234567891011

{  
  "\_id": "6abb75c5038864788971e17d",  
  "userId": "6abb6393ec7e6bccf1c5c344",  
  "action": "LOGIN",  
  "metadata": {  
    "description": "User logged in"  
  },  
  "createdAt": "2026-09-29T08:24:37.655Z",  
  "updatedAt": "2026-09-29T08:24:37.655Z",  
  "\_\_v": 0  
}

ConsoleNot connected to a Postman account

## Inventory Creation API

The screenshot shows the Postman interface for the 'Inventory Creation API'. The top navigation bar includes 'Home', 'Workspaces', and 'Explore'. A search bar is present. A notification banner at the top states: 'You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.' The left sidebar shows a 'History' panel with a list of requests. The main panel displays a POST request to 'http://127.0.0.1:8002/inventory'. The request body is a JSON object: 

```
{  "productId": "product_1",  "availableQuantity": 10}
```

. The response is a 200 OK status with a 52 ms response time and 194 B of data. The response body is a JSON object: 

```
{  "productId": "product_1",  "availableQuantity": 10,  "reservedQuantity": 0}
```

. The bottom status bar shows 'Console' and 'Not connected to a Postman account'.

## Inventory Retrieval API

The screenshot shows the Postman interface for the 'Inventory Retrieval API'. The top navigation bar includes 'Home', 'Workspaces', and 'Explore'. A search bar is present. A notification banner at the top states: 'You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.' The left sidebar shows a 'History' panel with a list of requests. The main panel displays a GET request to 'http://127.0.0.1:8002/inventory/product\_1'. The response is a 200 OK status with a 4 ms response time and 194 B of data. The response body is a JSON object: 

```
{  "productId": "product_1",  "availableQuantity": 10,  "reservedQuantity": 0}
```

. The bottom status bar shows 'Console' and 'Not connected to a Postman account'.

## Inventory Service Running

```
(.env) C:\Users\Vida Reddy\ecommerce-microservices\inventory-service>notepad app\routes.py

(.env) C:\Users\Vida Reddy\ecommerce-microservices\inventory-service>notepad app\main.py

(.env) C:\Users\Vida Reddy\ecommerce-microservices\inventory-service>notepad app\main.py

(.env) C:\Users\Vida Reddy\ecommerce-microservices\inventory-service>uvicorn app.main:app --reload --port 8002
INFO: Will watch for changes in these directories: ['C:\\Users\\Vida Reddy\\ecommerce-microservices\\inventory-service']
INFO: Uvicorn running on http://127.0.0.1:8002 (Press CTRL+C to quit)
INFO: Started reloader process [3104] using StatReload
INFO: Started server process [1892]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:64553 - "GET /inventory HTTP/1.1" 200 OK
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [1892]
INFO: Stopping reloader process [3104]

(.env) C:\Users\Vida Reddy\ecommerce-microservices\inventory-service>notepad app\database.py

(.env) C:\Users\Vida Reddy\ecommerce-microservices\inventory-service>notepad app\main.py

(.env) C:\Users\Vida Reddy\ecommerce-microservices\inventory-service>notepad app\routes.py

(.env) C:\Users\Vida Reddy\ecommerce-microservices\inventory-service>uvicorn app.main:app --reload --port 8002
INFO: Will watch for changes in these directories: ['C:\\Users\\Vida Reddy\\ecommerce-microservices\\inventory-service']
INFO: Uvicorn running on http://127.0.0.1:8002 (Press CTRL+C to quit)
INFO: Started reloader process [25224] using StatReload
INFO: Started server process [11160]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:64762 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:64762 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:51714 - "POST /inventory HTTP/1.1" 200 OK
```

## Product Creation API

The screenshot displays the Postman API client interface. At the top, there's a navigation bar with 'Home', 'Workspaces', and 'Explore' options, along with a search bar and 'Sign In'/'Create Account' buttons. A blue banner below the navigation bar states: 'You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.'

The main interface is divided into three sections:

- History:** A list of recent requests on the left, categorized by 'Today' and 'Yesterday'. Today's requests include POST to /products, GET to /product\_1, and POST to /inventory. Yesterday's requests include GET to /activities, POST to /activities, GET to /health, POST to /login, and POST to /users.
- Request Editor:** The central area shows a selected POST request to 'http://127.0.0.1:8001/products'. The 'Body' tab is active, displaying a JSON payload: 

```
{  "name": "Laptop",  "category": "Electronics",  "price": 50000}
```
- Response View:** The bottom section shows the response of the request. It includes a status bar indicating '201 Created', '25 ms', and '191 B'. The response body is displayed in the 'Pretty' tab, showing a JSON object: 

```
{  "id": "6abc9f925b5e70721782e47a",  "message": "Product created"}
```

At the bottom left, there's a 'Create collections in Postman' section with a 'Create a Collection' button. The bottom status bar shows 'Console' and 'Not connected to a Postman account'.

## Product Retrieval API

The screenshot shows the Postman interface with a GET request to `http://127.0.0.1:8001/products`. The response is a JSON array containing one product object. The product object has the following fields: `_id`, `name`, `category`, and `price`.

```
[{"_id": "6abc9f925b5e70721782e47a", "name": "Laptop", "category": "Electronics", "price": 50000.0}]
```

## User Registration API – Postman

The screenshot shows the Postman interface with a POST request to `http://127.0.0.1:4001/users`. The request body is a JSON object with the following fields: `name`, `email`, and `password`. The response is a JSON object representing a created user with the following fields: `id`, `name`, `email`, and `role`.

```
{ "name": "Test User", "email": "test@example.com", "password": "123456" }
```

```
{ "id": "6abca3f5ec7e6bccf1c5c346", "name": "Test User", "email": "test@example.com", "role": "CUSTOMER" }
```

## User Login API – Postman

The screenshot shows the Postman interface. The top bar includes navigation links (Home, Workspaces, Explore), a search bar, and user options (Sign In, Create Account). A notification banner at the top states: "You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman." The left sidebar displays a history of requests, categorized by "Today" and "Yesterday". The main workspace shows a POST request to `http://127.0.0.1:4001/login`. The request body is a JSON object: `{ "email": "test@example.com", "password": "123456" }`. The response tab is active, showing a 200 OK status with a response body containing a token: `{ "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiI2YWJjYTNmNWVjN2U2YmNjZjFjNWZlLCJyb2x1IjoiaQ1VTVE5NRVlCLCJpYXQiOiJlZ30TA3NDc4MTQsImV4cCI6MTc5MDc1MTQxNH0uYCCcoLAS11A20HX4316c6oNwKwKCAq0R6WClYFmIbQc" }`. The bottom status bar indicates "Console" and "Not connected to a Postman account".

## Product Service Health Check

The screenshot shows the Postman interface for a GET request to `/health`. The "Parameters" tab is active, showing "No parameters". Below the parameters, there are "Execute" and "Clear" buttons. The "Responses" tab is active, showing the "Curl" command: `curl -X 'GET' \ 'http://127.0.0.1:8003/health' \ -H 'accept: */*'` . The "Request URL" is `http://127.0.0.1:8003/health`. The "Server response" section shows a 200 status code and a response body: `{ "service": "product-service", "status": "UP" }`. A "Download" button is visible next to the response body.